

Drawings and PCB Photos - Automated Part Costing

Problem Statement

PII:

No

- Costing a circuit board means listing every component on it and pricing each one. A medium-complexity board takes two to three weeks.
- It needs someone who knows what electronics cost — a small capacitor, a six-layer board, placing a fine-pitch chip. Few people have that, so boards wait for them.
- On mechanical parts, most of what we need is written on the 2D drawing, not held in the 3D model, and is read by eye and typed in by hand.
- The costing method and the rate data do not change. Only the reading changes.

Projected ROI

Cost: £???

Return: £???

What is the return based on?

The boards we turn away, and the weeks a specialist spends on each one.

Time (hrs/week)

2-3 weeks per board

Current State

- For a board, someone identifies every component from a photo or a sample — package, value, rating, and the exact part number for each chip — then finds a price for every line at the volume we are asking about.
- The bare board is estimated separately for size, layers, finish and test, and then the assembly on top: how many placements, how many are fine-pitch or BGA, one side or two.
- All of it needs an electronics cost specialist, and two to three weeks. A design change means starting again.
- For a machined part, the drawing is read by eye and the tolerances, finish and heat treatment typed into CAPPe by hand.

Proposed Solution/Model

- Conduct a 3-stage Proof of Value (PoV), run the same way as the CAD to Cost PoV, and validate outputs vs existing cost estimates.
- For a board: upload a photo, say how many you are making and where it is built. The tool lists the components, prices each line, works out the bare board and the assembly, and gives the full cost.
- For a machined part: the drawing is read alongside the model, for the tolerances, finish and heat treatment the model cannot carry.
- Everything it reads is checked against a price catalogue or the measured geometry before it is used. It does not set any price.

Expected Benefits:

- **No Specialist Needed:** A cost engineer can cost a board without being an electronics pricing specialist. The specialist checks the answer instead of building it.
- **Wider Coverage:** We can cost the boards and parts we turn away today, and re-cost them when the design changes.

Technical Scoping:

Enabling AI on an existing system?
(YES)

The reading is done inside CostVision. CAPPe is not changed.

Model used: Claude (Anthropic)

Sonnet and Haiku to read, Opus where a board needs a closer look.

Technical implementation/support required?
(YES)

It needs an approved route to the model. Option 2 runs with that route switched off. This one cannot.